Major Project Proposal: Netflix Dataset Analysis

1. Introduction to Dataset

The dataset under consideration is a Netflix dataset containing 7,789 records and 11 columns. It includes important details about Movies and TV Shows available on the platform. Key attributes include the title, director, cast, country of origin, release date, rating, duration, and type (genres). The dataset spans multiple years, with entries ranging from 2008 to 2021, covering content across diverse countries and genres.

2. Problem Statement

Netflix has become one of the most prominent global streaming platforms, continuously expanding its library with a mix of original productions and licensed content. However, with growing competition from platforms like Amazon Prime, Disney+, and regional OTT providers, Netflix must strategically analyze its content catalog to identify strengths, gaps, and opportunities. The specific problem to be addressed in this project is 'Content Trends Analysis for Strategic Recommendations'. The aim is to uncover how Netflix's content distribution (Movies vs. TV Shows, genres, and country contributions) has evolved over the years. This will enable the identification of key genres, audience preferences, and strategic insights into global content expansion.

3. Importance of the Problem Statement

Understanding Netflix's content trends is crucial for making data-driven business decisions. The analysis not only highlights the balance between Movies and TV Shows but also reveals popular genres and underrepresented categories. For a platform that serves diverse international audiences, country-wise contributions provide valuable insights into global representation and market penetration. By focusing on these content trends, Netflix can refine its strategy for content acquisition and production, ensuring that it caters to the right audience segments while staying competitive in the global OTT industry.

4. Objectives

- Analyze the distribution of Movies vs. TV Shows over the years.
- Identify the most common genres and how their popularity has changed.
- Compare country-wise contributions to Netflix's catalog.

5. Expected Outcomes

- A clear understanding of how Netflix's content strategy has evolved.
- Identification of top-performing genres and categories.
- Strategic recommendations on which content types Netflix should focus on in the future.

```python
# --------------------------------------------------------
# Netflix Dataset Analysis - Problem Statement 1
# Content Trends Analysis for Strategic Recommendations
# --------------------------------------------------------

# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Setup visualization style
sns.set(style="whitegrid")
plt.rcParams['figure.figsize'] = (12,6)
```

```python
# --------------------------------------------------------
# 2. Load Dataset
# --------------------------------------------------------
df = pd.read_csv("Netflix Dataset.csv")

# Quick look at dataset
print("Shape of Dataset:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nMissing Values:\n", df.isnull().sum())
```

```
Shape of Dataset: (7789, 11)

Columns: ['Show_Id', 'Category', 'Title', 'Director', 'Cast', 'Country', 'Rel

Missing Values:
 Show_Id            0
Category          0
Title             0
Director       2388
Cast            718
Country         507
Release_Date     10
Rating            7
Duration          0
Type              0
Description       0
dtype: int64
```

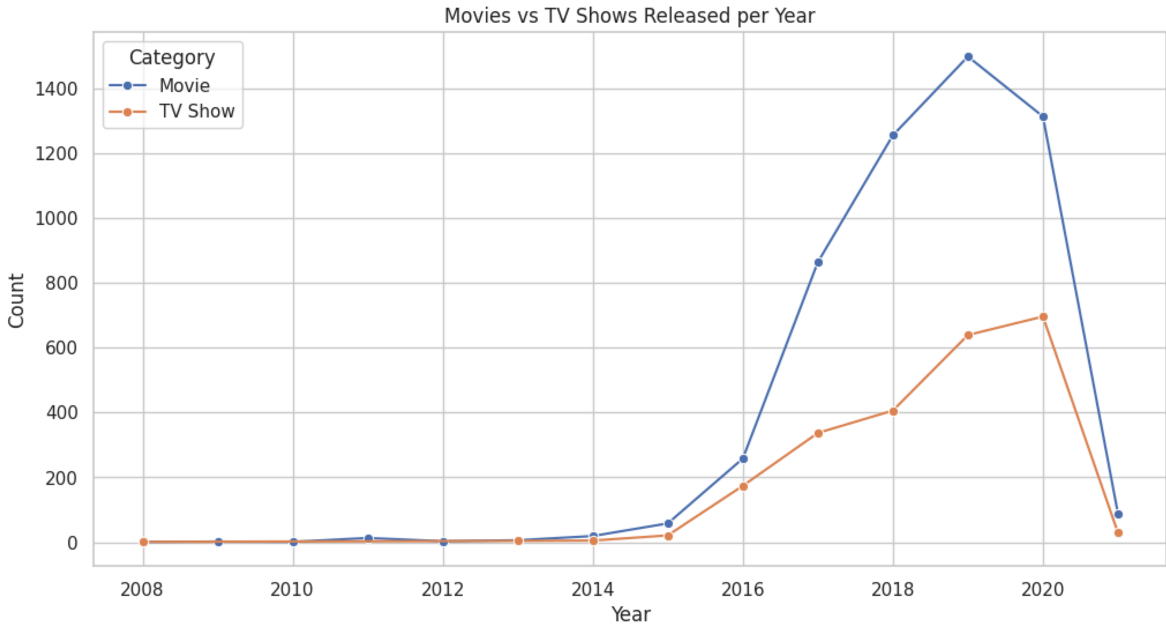
```python
# --------------------------------------------------------
# 3. Data Cleaning
# --------------------------------------------------------

# Convert Release_Date to datetime
df['Release_Date'] = pd.to_datetime(df['Release_Date'], errors='coerce')

# Extract Year
df['Year'] = df['Release_Date'].dt.year
```

```
# Handle missing values
df['Director'] = df['Director'].fillna("Unknown")
df['Cast'] = df['Cast'].fillna("Unknown")
df['Country'] = df['Country'].fillna("Unknown")
df['Rating'] = df['Rating'].fillna("Not Rated")

# Extract Main Genre (first genre listed in 'Type')
df['Main_Genre'] = df['Type'].apply(lambda x: x.split(",")[0].strip() if pd.
```

```
# -------------------------------------------------------
# 4. Exploratory Data Analysis (EDA)
# -------------------------------------------------------

# Count of Movies vs TV Shows
print("\nCategory Distribution:\n", df['Category'].value_counts())

# Top Countries
print("\nTop 5 Countries:\n", df['Country'].value_counts().head(5))

# Dataset span (years)
print("\nDataset covers years from", df['Year'].min(), "to", df['Year'].max(
```

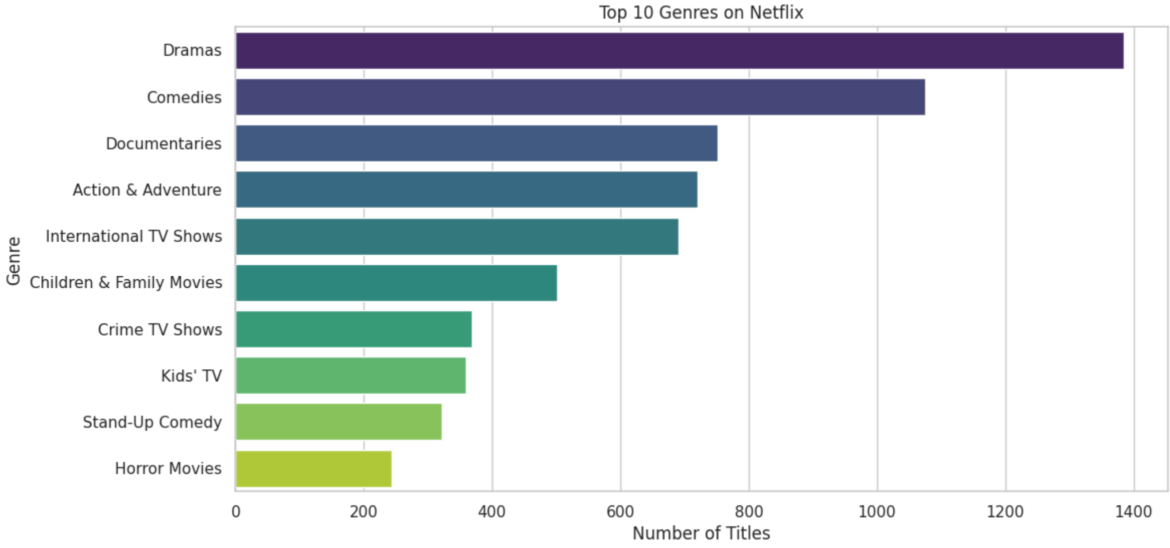
```
Category Distribution:
 Category
Movie      5379
TV Show    2410
Name: count, dtype: int64

Top 5 Countries:
 Country
United States     2556
India              923
Unknown            507
United Kingdom     397
Japan              226
Name: count, dtype: int64

Dataset covers years from 2008.0 to 2021.0
```

```
# ------------------------------------------------------
# 5. Analysis & Visualizations
# ------------------------------------------------------

# 1. Line Chart - Movies vs. TV Shows released per year
content_trend = df.groupby(['Year', 'Category']).size().reset_index(name='Cc
sns.lineplot(data=content_trend, x='Year', y='Count', hue='Category', marker
plt.title("Movies vs TV Shows Released per Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```

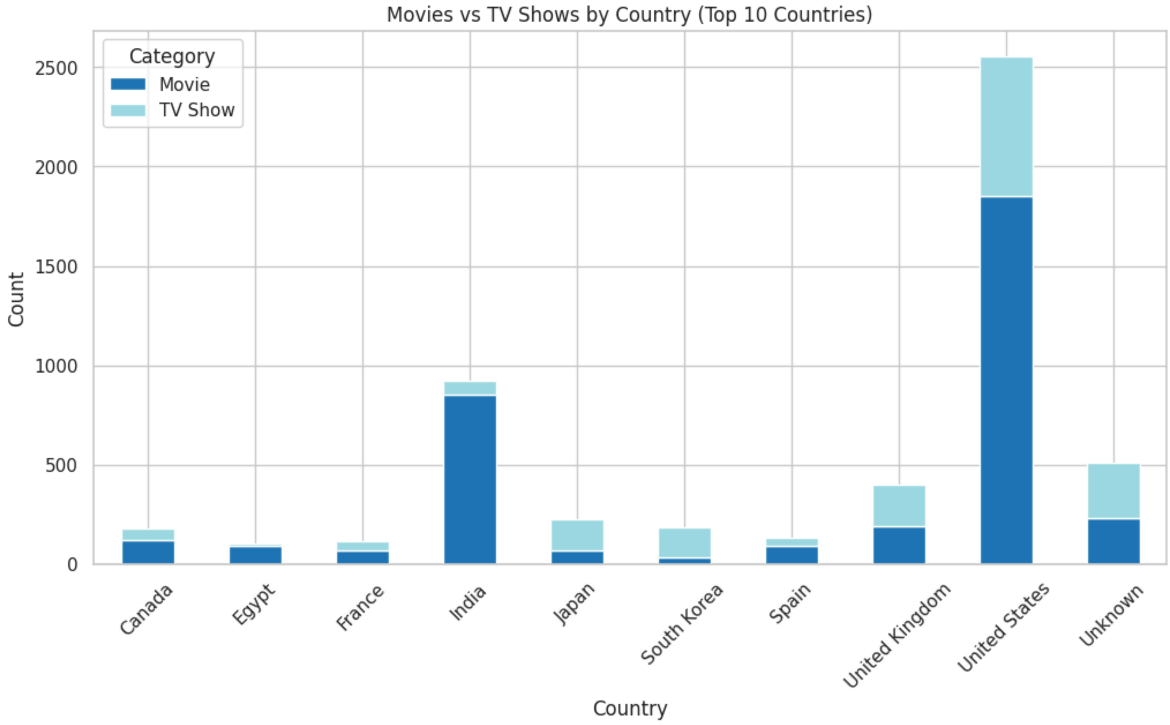



```
# 2. Bar Chart - Top 10 Genres
top_genres = df['Main_Genre'].value_counts().head(10)
sns.barplot(x=top_genres.values, y=top_genres.index, palette="viridis")
plt.title("Top 10 Genres on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.show()
```

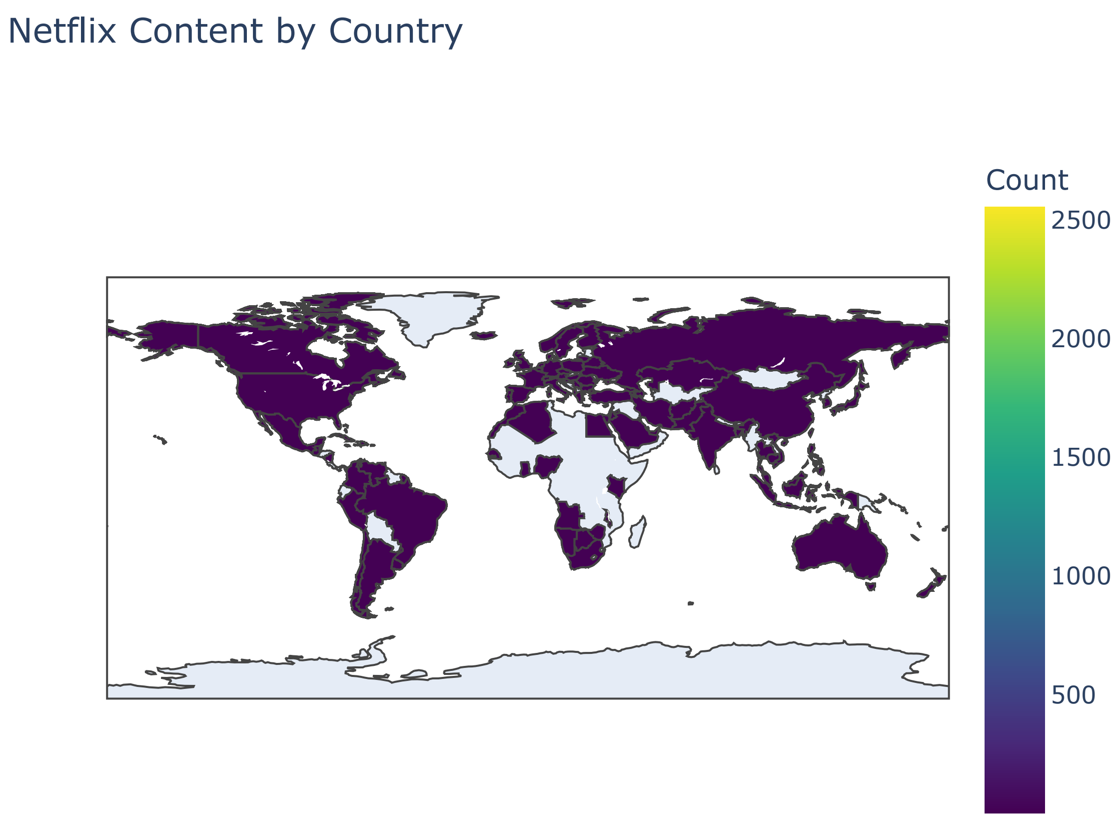
```
/tmp/ipython-input-76000803.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

  sns.barplot(x=top_genres.values, y=top_genres.index, palette="viridis")
```



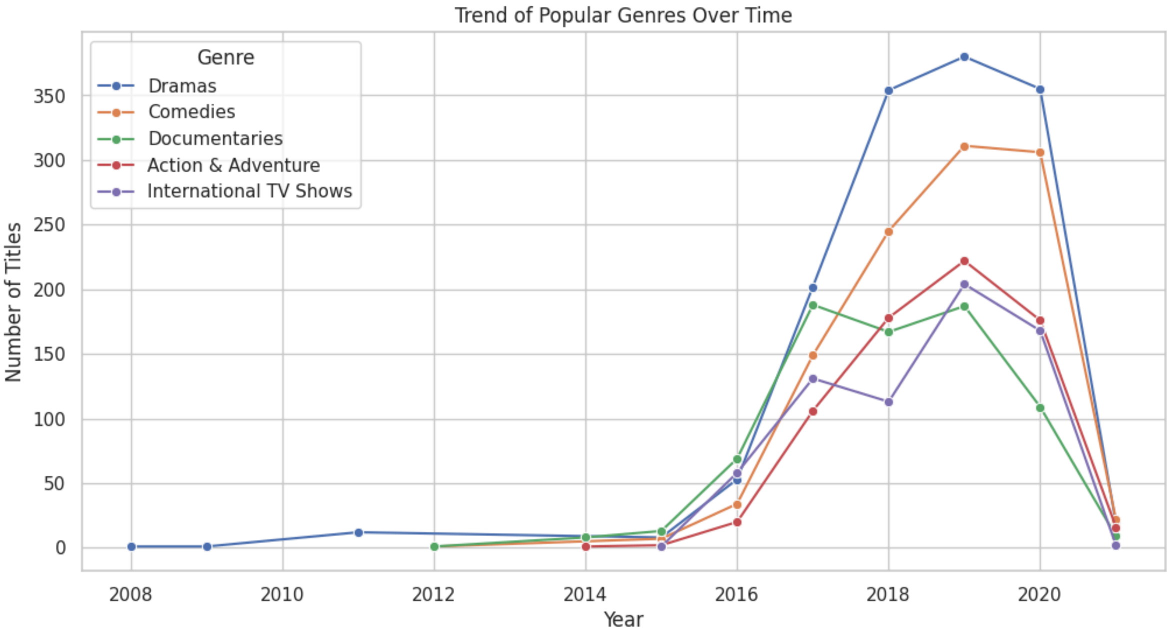

```
# 3. Stacked Bar Chart – Movies vs TV Shows per Country (Top 10 countries)
top_countries = df['Country'].value_counts().head(10).index
country_data = df[df['Country'].isin(top_countries)]
stacked = country_data.groupby(['Country', 'Category']).size().unstack(fill_
stacked.plot(kind='bar', stacked=True, figsize=(12,6), colormap="tab20")
plt.title("Movies vs TV Shows by Country (Top 10 Countries)")
plt.xlabel("Country")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```



```
# 4. Choropleth Map - Country-wise Content
country_counts = df['Country'].value_counts().reset_index()
country_counts.columns = ['Country', 'Count']
fig = px.choropleth(country_counts, locations="Country", locationmode="count
                    color="Count", hover_name="Country",
                    color_continuous_scale="viridis", title="Netflix Content
fig.show()
```




```
# 5. Trend Line - Growth of Popular Genres Over Time
popular_genres = df['Main_Genre'].value_counts().head(5).index
genre_trends = df[df['Main_Genre'].isin(popular_genres)].groupby(['Year','Ma

sns.lineplot(data=genre_trends, x='Year', y='Count', hue='Main_Genre', marke
plt.title("Trend of Popular Genres Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.legend(title="Genre")
plt.show()
```



Trend of Popular Genres Over Time
Genre
Dramas
Comedies
Documentaries
Action & Adventure
International TV Shows
Number of Titles
0
50
100
150
200
250
300
350
2008
2010
2012
2014
2016
2018
2020
Year